L7: Agglomerative Hierarchical Clustering Approaches - Node Similarity Metric

Let us now switch to agglomerative *(or bottom-up)* hierarchical clustering algorithms.

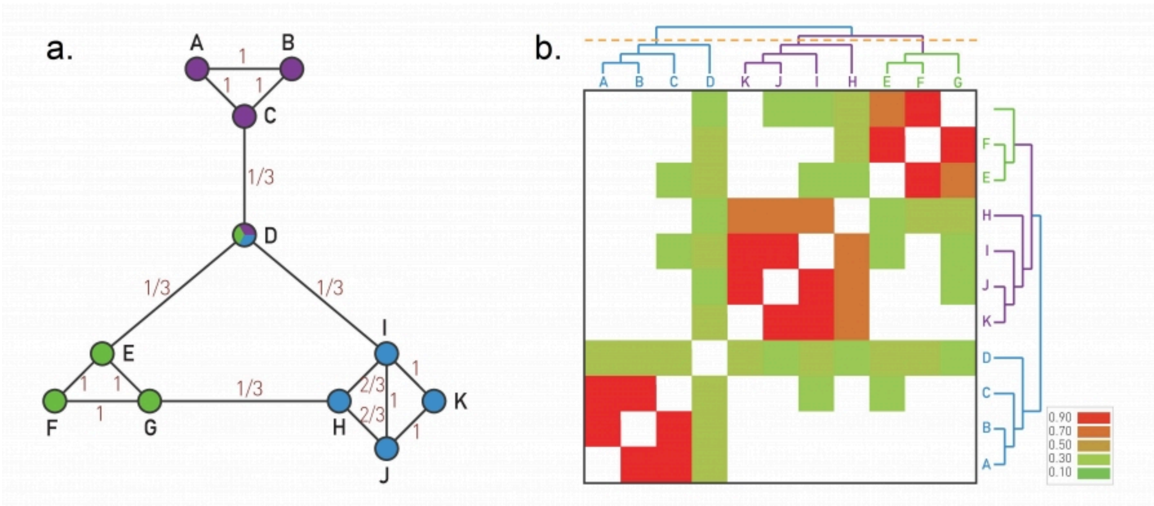

Image 9.9 from networksciencebook.com . The Ravasz Algorithm The agglomerative hierarchical clustering algorithm proposed by Ravasz was designed to identify functional modules in metabolic networks, but it can be applied to arbitrary networks.

Here, we start the dendrogram at the leaves, one for each node. In each iteration, we decide which nodes to merge so that we extend the dendrogram by one branching point towards the top. The merged nodes should ideally belong to the same community. So we first need a metric that quantifies how likely it is that two nodes belong to the same community.

If two nodes, *i* and *j*, belong to the same community, we expect that they will both be connected to several other nodes of the same community. So, we expect that *i* and *j* have a large number of common neighbors, relative to their degree.

To formalize this intuition, we can define a similarity metric $S_{i,j}$ between any pair of nodes *i* and *j*:

$$S_{i,j} = \frac{N_{i,j}+A_{i,j}}{\min\{k_i,k_j\}}$$

where $N_{i,j}$ is the number of common neighbors of *i* and *j*, $A_{i,j}$ is the adjacency matrix element for the two nodes (1 if they are connected, 0 otherwise), and k_i is the degree of node *i*.

Note that $S_{i,j} = 1$ if the two nodes are connected with each other and every neighbor of the lower-degree node is also a neighbor of the other node.

On the other hand, $S_{i,j} = 0$ if the two nodes are not connected to each other and they do not have any common neighbor.

The visualization at the left shows the node similarity value for each pair of connected nodes.

The visualization at the rights shows the color-coded node similarity matrix, for every pair of nodes *(even if they are not connected).* Note that three groups of nodes emerge with higher similarity values: *(A, B, C), (H,I,J,K)* and *(E,F,G).* Node D on the other hand has a lower similarity with any other node, and it appears to be a ***"connector"*** between the three communities.

Food For Thought

What if a node has no connections? How should we modify this similarity metric to deal with that case?